E2E Automation Project:

Homburg Appointment System

Project Overview & Test Plan

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Technologies: Playwright, JavaScript, GitHub

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1. Project Overview & Scope

- **System Under Test (SUT):** termine-reservieren.de/termine/homburg/.
- **Objective:** Designing an automated framework using Playwright.
- **Safety Policy:** Submitting data at Step 6 is strictly prohibited to prevent database pollution in the government institution's system.

2. Testing Approaches

- **Manual Testing:** Exploratory, Functional, UI/UX Review, Accessibility Audit (BITV 2.0 / WCAG 2.1).
- **Automation Testing:** Functional E2E Testing via Playwright.

3. Test Environment & Stack

- **Browsers:** Chromium (Chrome/Edge), Firefox, WebKit (Safari).
- **Devices:** Desktop (1920x1080), Mobile (iPhone 13 / 430px).
- **Technical Stack:** Playwright, JavaScript, Node.js, Git/GitHub.

4. Test Coverage (Key Modules)

- **Accessibility:** Audit of keyboard navigation (Tab, Enter, Space) and screen reader compatibility.
- **UI Components:** Interactive accordions and dynamic category lists.
- **Forms:** Input validation, data masking, and E-mail field protection.
- **Integrations:** Maps, geodata, and external navigation.
- **Session Logic:** Persistent countdown timers and predictive info blocks.

5. Out of Scope

External links, backend, database, security, and load testing.

Project Motivation & Rationale

Why this project? The choice of the termine-reservieren.de booking system is driven by the need to test a real-world, high-demand service that residents in Germany interact with daily. Working with this system provided not only a user-centric perspective but also the opportunity to perform an in-depth analysis of logical interface vulnerabilities.

Technical Approach Leveraging my background in manual testing and foundational knowledge of JS and OOP, I designed an automated Playwright framework tailored for a complex SPA - Single Page Application. The project focuses on covering the following critical mechanics:

- **Accessibility (a11y):** Manual audit (BITV 2.0 / WCAG 2.1), testing keyboard navigation, and ensuring screen reader compatibility.
- **UI Components:** Testing dynamic accordions, interactive maps (geotags, pop-ups), session elements (timers, predictive info blocks), slot selection filters, and multi-step registration forms.
- **Form Validation:** Validating complex inputs, handling data masks, and securing E-mail fields.

Key Skills Demonstrated Implementing this project allowed me to demonstrate proficiency in building robust, anti-flaky test logic, managing session states, utilizing API mocking, and applying deep knowledge of modern software accessibility standards.

Test Plan

Software Testing Lifecycle (STLC)

- **I. Requirements Analysis:** Analyzing system logic, breaking down requirements into User Stories, and defining Acceptance Criteria using the BDD approach (Given-When-Then).
- **II. Manual Testing:** Developing detailed test cases ready for automation and checklists for accessibility (BITV 2.0). Creating a Requirements Traceability Matrix (RTM) for 100% test coverage.
- **III. Test Automation Framework Engineering:** Designing a scalable Playwright + JavaScript framework architecture. Developing stable E2E tests for positive, negative, and system scenarios, including API Mocking.
- **IV. CI/CD Integration, Execution & Reporting:** Local and pipeline test execution. Gathering quality metrics and generating comprehensive reports. Conducting Root Cause Analysis (RCA) using logs, screenshots, and Playwright Trace Viewer.
- **V. Maintenance & Test Flakiness Management:** Maintaining test documentation and updating the codebase in response to UI/UX changes. Optimizing locator stability and wait strategies to ensure high test reliability.

I. Requirements & Design

- **1.1 End-to-End User Flow & Step-by-Step Walkthrough:** Detailed step-by-step description of the user's primary journey (Happy Path) from Step 1 through Step 6.
- **1.2 Technical & Interface Specifications:** Technical rules covering session limits, field validation, UI states, and UX standards.
- **1.3 User Personas & Roles:** Defined user profiles for conducting exploratory testing.
- **1.4 Functional Requirements (User Stories):** User stories detailing all system functionalities.
- **1.5 Acceptance Criteria (Given-When-Then):** Acceptance criteria defining the logic for each user story using the BDD format.
- **1.6 Test Scenarios (Positive & Negative):** Segmentation of test cases into positive (Happy Path) and negative (Edge Cases) scenarios.

1.1 End-to-End User Flow & Step-by-Step Walkthrough

- **Step 1: Functional Unit Selection**
  - **Elements:** Bürgeramt (Citizens' Office), Fahrerlaubnis (Driver's License).
  - **Action:** Selecting the relevant department.
- **Step 2: Service Selection**
  - **Logic:** Selecting services (maximum of 4 combined).
  - **Navigation:** "Zurück zu den Funktionseinheiten" (Back to functional units) and "Weiter" (Continue) buttons.
  - **Overlay:** "Hinweis" (Notice) window with "Ok", "Abrechnen", and "X" (Close) buttons.
- **Step 3: Branch Selection**
  - **Action:** Selecting a branch office via an interactive map.
  - **Navigation:** "Zurück zu den Anliegen" (Back to services) and "Weiter" (Continue) buttons.
- **Step 4: Time Slot Selection**
  - **Action:** Selecting a date from the calendar and a specific time slot; filtering functionality is available.
  - **Navigation:** "Zurück zu den Standorten" (Back to locations) button.
  - **Overlay:** Confirmation window with "Ja" (Yes), "Nein" (No), and "X" (Close) buttons.
- **Step 5: Personal Data Entry**
  - **Required Fields (*) for Bürgeramt:** Vorname (First name), Nachname (Last name), E-Mail, E-Mail-Wiederholung (Email confirmation), Geburtsdatum (Date of birth - year sufficient), and "Ich willige ein" (Consent) checkbox.
  - **Required Fields (*) for Fahrerlaubnis:** Vorname, Nachname, E-Mail, E-Mail-Wiederholung, Geburtsdatum, Postleitzahl (Postal code), Wohnort (City), and "Ich willige ein" (Consent) checkbox.
  - **Optional Fields:** Telefonnummer (Phone number), Straße (Street), Hausnummer (House number).
  - **Navigation:** "Zurück zu den Terminvorschlägen" (Back to appointment suggestions) and "Reservieren" (Reserve) buttons.
- **Step 6: Confirmation (Out of Scope)**
  - **Result:** Final screen confirming the reservation.
  - **Isolation:** Email confirmation link delivery.

1.2 Technical & Interface Specifications

1.2.1 Constraints & Rules

- **Sequential Workflow:** Steps must be navigated strictly using "Back" / "Continue" buttons; direct URL access is restricted and triggers an error.
- **Global Service Limit:** Maximum of 4 services (Anliegen) allowed per booking session.
- **Multi-Category Restriction:** Selecting services from different departments in one session is prohibited; changing the department at Step 1 clears the selected services cache.
- **Counter Logic:** The minus (-) button is disabled when the value is 0. The plus (+) button is disabled for all services in expanded accordions once the global limit of 4 services is reached.

- **Read-Only Summary:** The "Übersicht" panel is an expanded-by-default accordion displaying selected data in read-only mode; data editing within this panel is disabled.
- **Session Integrity:** Simultaneous operation in multiple browser tabs is prohibited; parallel booking attempts return a session error.
- **Single Slot Selection:** Selecting a new time slot automatically cancels any previously selected slot.
- **Reservation Hold:** Selected slots are held for the user for the duration of the 25-minute session timer.
- **Data Privacy Consent:** The "Reservieren" (Reserve) button remains disabled until the user activates the data processing consent checkbox.
- **Email Lifecycle Rule:** Booking confirmation is completed only after clicking the link provided via Email; this link is also used for cancellation (Stornieren) or rescheduling (Umbuchen).

1.2.2 Input & Validation

- **Security & Base Validation:**
  - **XSS/Injection:** HTML tags and script tags (<script>) are automatically blocked in all input fields.
  - **Trim:** Excess whitespace is automatically removed from the beginning and end of all text fields before data submission.
  - **Mandatory Indicators:** Fields required for submission are marked with an asterisk (*).
- **Field Specifications:**
  - **Name Fields (Vorname/Nachname):** Text fields (max 200 chars); letters (including umlauts), hyphens, apostrophes, and spaces allowed; special characters and digits prohibited.
  - **Email Validation:** Validated against the name@domain.tld mask. "E-Mail" and "E-Mail-Wiederholung" fields must match exactly. Paste functionality is blocked for the confirmation field.
  - **Phone Number (Telefon):** Optional field (max 20 chars); digits only.
  - **Date of Birth (Geburtsdatum):** Only the validity of the birth year is verified.
  - **Address Fields (Straße/Hausnummer):** Optional fields allowing both letters and digits.
  - **Postleitzahl/Wohnort (Fahrerlaubnis):** Mandatory for driver's license applications. Postal code accepts exactly 4 or 5 digits; city requires a minimum of 3 characters.

1.2.3 UI States & Handling

1. **Header Controls**
   - **Header Icons (Contrast, Einfache Sprache / Language Selection):**
     - **Default:** Elements are interactive (pointer cursor), transparent background, grey icon color.
     - **Hover:** Visual inversion of icon and background style upon hover.
     - **Dropdown (Click on 2nd icon):** Activates dropdown menu with "Deutsch" link.
   - **Logo (City Logo):**
     - **Static:** Non-interactive image.

2. **Action Buttons & Navigation**
    - **Active:** Dark green fill, white text, pointer cursor.
    - **Hover:** Programmatic opacity reduction, resulting in a faded visual appearance.
    - **Disabled:** Triggered when step conditions are unmet; semi-transparent light green fill, grey text, not-allowed cursor.
3. **Accordions & Summary Panel**
    - **Service Categories:** Interactive by default (pointer cursor); toggles content visibility (Expanded / Collapsed).
    - **Filter & Time Selection:** Accordion logic implemented for filters; filtering synchronized with calendar view.
    - **Übersicht Panel (Summary):** Interactive header for toggling visibility; content is read-only with editing disabled.
4. **Counter Controls (Step 2)**
    - **Button (-):** Disabled when value is 0 (not-allowed cursor); Active when value $\geq$ 1 (pointer cursor).
    - **Button (+):** Active by default; Disabled when the global limit (4 services total) is reached.
5. **Visual Slot States (Step 4)**
    - **Available:** Interactive state (pointer cursor).
    - **Unavailable:** Deactivated state; booking blocked; hover and click events inactive.
6. **Validation Status Indicators**
    - **Error Marker [!]:** Triggered on blur if validation fails; static red triangle icon.
    - **Success Marker [✓]:** Triggered on blur upon successful validation; static green circle icon.
    - **Tooltips (?):** Grey circular icon; triggers overlay text on click.
7. **Modals & Overlays**
    - **Modal Trigger:** Blocking overlay prevents interaction with the background interface.
    - **Interactive Elements:** Primary action buttons ("Ok", "Schliessen") feature green fill with hover-opacity effects; secondary buttons and close [X] icon are interactive.
8. **Session Timer**
    - **Static Text:** Non-interactive element.
    - **Dynamic State:** String-based object updating every minute (countdown from 25 to 00).
9. **Inline & Footer Links**
    - **Default:** Green text with underline.
    - **Hover:** Opacity reduction.
    - **Overlay Trigger (Footer):** Activates modal window without page reload.
    - **Target Isolation:** Uses target="_blank" attribute for external document links.
10. **Checkboxes (Datenschutz Step 5)**
    - **Unchecked:** Default state; empty box.
    - **Checked:** User-activated trigger; validation passed.

1.2.4 Session & Navigation

- **Session Lifecycle:**
    - **Session Initiation:** The session is activated upon loading Step 1; a global 25-minute countdown timer is enabled in the page footer.
    - **Data Persistence:** Selected parameters (services, location) are maintained between steps 1–3 during navigation via "Back" / "Continue" buttons and page refreshes (F5).
    - **Temporary Slot Reservation:** Upon confirming a time slot at Step 4, the slot is temporarily locked to the session until the process is completed or the timer expires.
    - **Data Storage & Privacy:** Data entered at Step 5 is stored in session memory (client-side) and is automatically destroyed upon closing the browser tab.
- **Expiration Handling:**
    - **Visual Timer:** Persistent countdown timer displayed across all UI layers.
    - **Soft Timeout (Warning Overlay):** At 01 remaining, a blocking modal appears, allowing the user to extend the session to 25 by clicking "Close".
    - **Hard Timeout (Termination):** If no action is taken by 00:00, the session is annulled, the booking slot is released, and the user is redirected to Step 1.
- **Navigation Logic & Resilience:**
    - **Browser Back Button:** Pressing the browser "Back" button does not reset the session and returns the user to the previous step; data cached within the session is re-validated upon return from Step 5.
    - **Progress Tracking:** The progress bar accurately displays the current stage across all 6 steps.
- **System Integrity:**
    - **SSL/HTTPS Encryption:** All data transmission processes are protected by the TLS (HTTPS) encryption protocol.
    - **Offline Handling:** In the event of internet connection loss during transitions, the site displays a system notification ("Check connection") without immediately resetting session data.

1.2.5 Inclusivity, UX & GDPR

- **Accessibility (BITV 2.0 / WCAG 2.1 Compliance):**
    - **Keyboard Navigation:** All interactive elements (buttons, slots, checkboxes) are accessible via the Tab key and activatable using Enter or Space.
    - **Accessibility Handling:** All interactive and graphical interface elements utilize technical aria-label attributes to ensure correct interpretation by screen readers.
    - **State Reporting:** Dynamic components (accordions, validation markers) communicate current system status via specialized ARIA properties.
    - **Focus Indication:** A clear, visible focus ring surrounds the active element and remains present when modal windows are opened.
    - **Contrast Toggle:** A high-contrast mode (Black/White) is available and persists throughout all step transitions.
    - **Color Contrast:** All text and icons maintain a minimum contrast ratio of 4.5:1.
- **Responsive Behavior (Mobile Adaptability):**
    - **Touch Targets:** The minimum touch area for mobile elements is 44x44 pixels.

    - **Adaptive Layout:** For screen widths under 480px, the calendar and service lists reconfigure into a single vertical column; horizontal scrolling is strictly prohibited.
    - **Mobile Summary View:** The service summary view is collapsible on smaller resolutions to ensure it does not obstruct primary fields.
- **User Experience & Feedback:**
    - **Visual Hierarchy:** Primary action buttons ("Weiter", "Reservieren") are visually dominant, utilizing high-contrast brand colors.
    - **Language Consistency:** The platform includes an **"Einfache Sprache an"** (Plain Language) toggle at the top of the interface, providing an accessible version of the content for users with varying language proficiency or cognitive needs.
- **Privacy & GDPR (DSGVO):**
    - **Data Minimization:** The system requests only the minimum dataset required for identification.
    - **Privacy Links:** Direct links to "Impressum" and "Datenschutz" are available in the footer at every stage to ensure compliance with German law.

1.3 User Personas & Roles

1. **Bürger / Standard User:** A user booking an appointment independently via desktop or mobile device. The primary goal is to successfully navigate all steps, secure a time slot, and validate the email address. This user expects the session timer to function accurately and requires clear error guidance when inputting data.
2. **Accessible User:** A user with accessibility needs who navigates the interface solely via keyboard (Tab, Enter, Space) or uses screen readers. This persona verifies strict compliance with the BITV 2.0 standard.
3. **Sachbearbeiter / Clerk (Out of Scope):** An administrative employee processing confirmed bookings within the internal CRM/Backend system. This role is outside the scope of UI Black Box testing.

1.4 Functional Requirements (User Stories)

Role: Bürger (Standard User)

- **US_01: Entry Point & Session Initialization (Pre-step)**
    - **Statement:** As a User, I want to navigate to the booking page via a direct URL so that I can begin the appointment process from the first step.
- **US_02: Department Selection (Step 1)**
    - **Statement:** As a User, I want to select a specific department (Bürgeramt or Fahrerlaubnis) so that I can immediately view the relevant services.
- **US_03: Multi-Service Selection (Step 2)**
    - **Statement:** As a User, I want to select multiple services simultaneously (up to 4) and view related tooltips so that I can manage all tasks in a single visit.
- **US_04: Required Documents Information (Step 2 - Hinweis)**
    - **Statement:** As a User, I want to view a list of required documents so that I can fully prepare for my appointment.
- **US_05: Location Selection (Step 3)**

- **Statement:** As a User, I want to view available branches on an interactive map so that I can select a suitable location and proceed to the calendar.
- **US_06: Time Slot Selection (Step 4)**
  - **Statement:** As a User, I want to view an interactive calendar and filtering tools so that I can quickly find and secure a convenient appointment time.
- **US_07: Data Entry & Form Submission (Step 5)**
  - **Statement:** As a User, I want to complete the form with my contact details and submit the request so that the institution can register my appointment.
- **US_08: Booking Confirmation (Step 6)**
  - **Statement:** As a User, I want to receive an on-screen booking confirmation and an activation link via email so that I can complete the registration process.
  - **Status:** Out of Scope.
  - **Rationale:** Testing this stage requires performing a real database transaction, which would pollute the production environment with invalid records.
- **US_09: Data Privacy (GDPR / Datenschutz)**
  - **Statement:** As a User, I want to see a mandatory consent checkbox for data processing at Step 5 and have access to the privacy policy so that my rights are protected in compliance with GDPR.

Role: User with Accessibility Needs

- **US_10: Interface Accessibility (Accessibility - BITV 2.0)**
  - **Statement:** As a User utilizing keyboard navigation or screen reader software, I want all interactive elements and system states to be semantically marked up so that I can independently complete the booking without using a mouse.

System Resilience & Non-Functional Requirements

- **SYS_01**: Session Timeout Warning (01:00)
- **SYS_02**: Session Expiration & Hard Timeout (00:00)
- **SYS_03**: Network Failure Handling (Offline Simulation)
- **SYS_04**: Parallel Booking Conflict Handling
- **SYS_05**: Backend Failure / API 500 Error Handling

Infrastructure & Responsiveness Requirements

- **INF_01**: Mobile Layout Adaptability (Viewport 430px)
- **INF_02**: Touch Target Sizing (44x44px)
- **INF_03**: Modal Overlay Isolation & Keyboard Trapping

1.5 Acceptance Criteria (BDD: Given-When-Then)

US_01: Entry Point & Session Initialization

- **AC_01.1: Successful Page Load**

- Given: The user has the direct URL to the booking system.
    - **When:** The user navigates to `https://termine-reservieren.de/termine/homburg/`.
    - **Then:** Step 1 page loads successfully, displaying the correct department selection interface.
- **AC_01.2: Deep Linking Protection**
    - **Given:** The user does not have an active booking session (cleared cache or new browser tab).
    - **When:** The user attempts to access an internal step URL directly (e.g., Step 2 URL).
    - **Then:** The system blocks access and displays an error page informing the user that direct navigation is not permitted.

US_02: Department Selection (Step 1)

- **AC_02.1: Automatic Navigation to Service List**
    - **Given:** The user is on the "Schritt 1 von 6" (Step 1 of 6) page.
    - **When:** The user clicks on either the "Bürgeramt" or "Fahrerlaubnis" button.
    - **Then:** The system redirects the user to "Schritt 2 von 6" (Step 2 of 6).

US_03: Multi-Service Selection (Step 2)

- **AC_03.1: Accordion Behavior**
    - **Given:** The user is on the "Schritt 2 von 6" (Step 2 of 6) page, and the list of service categories is displayed.
    - **When:** The user clicks on any category row.
    - **Then:** The category expands to display the nested list of services, tooltip icons (?), +/- buttons, and selection counters. Clicking again collapses the category, hiding the content.
- **AC_03.2: Counter Plus & Navigation Activation**
    - **Given:** All service counters are set to 0, and the "Weiter" (Continue) button is disabled.
    - **When:** The user clicks the (+) button for any service.
    - **Then:** The counter for that service increases to 1, the (-) button becomes active, and the "Weiter" button transitions to the enabled state.
- **AC_03.3: Counter Minus & Navigation State**
    - **Given:** A service is selected (counter equals 1).
    - **When:** The user clicks the (-) button.
    - **Then:** The counter decreases to 0, and the (-) button becomes disabled. If the total counter for all services on the page becomes 0, the "Weiter" button is automatically disabled.
- **AC_03.4: Service Limit Enforcement (Lock)**
    - **Given:** The user has selected 3 services.
    - **When:** The user clicks the (+) button on any service.
    - **Then:** All (+) buttons on the page transition to the disabled state. The (-) buttons for selected services and the "Weiter" button remain active.
- **AC_03.5: Service Limit Enforcement (Unlock)**
    - **Given:** The user has selected 4 services from any combination of categories.

- **When:** The user clicks the (-) button on any selected service.
- **Then:** All (+) buttons on the page transition to the active state. The (-) buttons for selected services and the "Weiter" button remain active.

US_04: Required Documents Information (Step 2 - Hinweis)

- **AC_04.1: Overlay Rendering**
  - **Given:** The user has selected between 1 and 4 services on Step 2.
  - **When:** The user clicks the "Weiter" (Continue) button.
  - **Then:** A modal window opens displaying the list of required documents and control elements: [X] (close), "Ok" (proceed with registration), and "Abrechnen" (cancel/close).

US_05: Location Selection (Step 3)

- **AC_05.1: Branch Display & Map Interaction**
  - **Given:** The user is provided with a choice of branches on the map.
  - **When:** The user clicks the "Karte anzeigen" (Show map) button on a branch card.
  - **Then:** An interactive map with branch markers/pins is rendered below the card, enabled for panning and zooming.
- **AC_05.2: Popup Activation**
  - **Given:** The user is provided with a choice of branches on the map.
  - **When:** The user clicks on a blue branch marker-pin on the map.
  - **Then:** An information window opens with "Standort auswählen" (Select location) and "Hierhin navigieren" (Navigate here) buttons.
- **AC_05.3: Branch Selection**
  - **Scenario 1 (Selecting Location):** Given the info window is active, when clicking "Standort auswählen", then the location is saved and the user is redirected to Step 4.
  - **Scenario 2 (Navigation):** Given the info window is active, when clicking "Hierhin navigieren", then Google Maps opens in a new tab (`target="_blank"`), leaving the session active.

US_06: Time Slot Selection (Step 4)

- **AC_06.1: 3-Week Horizon Availability**
  - **Given:** The user navigates to Step 4.
  - **When:** The page loads completely, and the system renders the list of available days for a 3-week period.
  - **Then:** The first available day is automatically expanded, displaying the grid of active time slots (time tiles).
- **AC_06.2: Filter Interaction (Accordion Filter Behavior)**
  - **Given:** The "Vorschläge filtern" (Filter suggestions) panel is in a collapsed state.
  - **When:** Clicking "Vorschläge filtern", it expands with "Zeitbereich" and "Wochentag" accordions.
  - **Scenario 1 (Filtering):** Selecting parameters and clicking "Filtern" updates the calendar grid.

    - **Scenario 2 (Reset):** Clicking "Zurücksetzen" resets filters to defaults and updates the grid.
- **AC_06.3: No Available Slots Error Scenario**
    - **Given:** The user applies filter criteria returning zero available slots.
    - **When:** The filter query returns an empty result.
    - **Then:** The calendar grid is hidden, an error message "Kein freier Termin verfügbar..." (No available appointments) is displayed, the "Weiter" button is disabled, and only the "Zurück zu den Standorten" button remains active.
- **AC_06.4: Time Selection Overlay Trigger**
    - **Given:** The user is on the Step 4 page.
    - **When:** The user clicks on an available dark-green time slot.
    - **Then:** A "Hinweis" modal window opens, blocking the main interface.
    - **Then:** A "Hinweis" modal opens. Clicking "Ja" secures the slot and proceeds to Step 5; clicking "Nein" or [X] closes the modal without saving.

US_07: Data Entry & Form Submission (Step 5)

- **AC_07.1: Global Validation UI Mechanics**
    - **Given:** Step 5 page is loaded, and all mandatory fields are marked with an error indicator [!].
    - **When:** The user fills in a field with valid data and moves the focus (blur event).
    - **Then:** The error marker [!] instantly changes to the success marker [✓]. If the field data is completely deleted, the error marker returns.
- **AC_07.2: Anti-Paste Restriction & Mismatch**
    - **Given:** The "E-Mail" field contains a valid address.
    - **When:** The user attempts to paste text into "E-Mail-Wiederholung".
    - **Then:** The interface blocks the paste action. The success marker [✓] in the "E-Mail-Wiederholung" field activates only when the text is entered manually and matches the text in the first field exactly.
- **AC_07.3: Date of Birth Validation Logic (Geburtsdatum)**
    - **Given:** Date of birth fields are empty.
    - **When:** Leaving Tag/Monat empty and entering a valid 4-digit year in "Jahr".
    - **Then:** The system activates the success marker [✓] for the year input.
- **AC_07.4: Submit Button Lock State**
    - **Given:** The form is displayed.
    - **When:** Any mandatory field or checkbox displays an error marker [!].
    - **Then:** The "Reservieren" (Reserve) button is in a disabled state.
- **AC_07.5: Submit Button Activation**
    - **Given:** All mandatory fields are validly filled and consent is checked.
    - **When:** All error markers transition to green checkmarks [✓] .
    - **Then:** The "Reservieren" button transitions to the active state.
- **AC_07.6: Backward Navigation Cache Retention**
    - **Given:** The user has reached Step 5.
    - **When:** Navigating backward using the "Zurück" button to steps 4, 3, or 2.
    - **Then:** Previously selected services, location, and time slot parameters remain cached and populated in the UI.
- **AC_07.7: Dynamic Übersicht & Step Indicator Synchronization**

    - **Given:** The user navigates through any step of the booking process.
    - **When:** The user advances or goes back a step.
    - **Then:** The Übersicht panel and progress indicator update dynamically to accurately reflect current choices and active step status.

US_08: Booking Confirmation (Step 6) Status: Out of Scope (Testing this stage requires real database transactions in the live city portal).

US_09: Data Privacy Compliance (GDPR / Datenschutz)

- **AC_09.1: Interface Blocking**
    - **Given:** The user is on the Step 5 page.
    - **When:** The consent checkbox is unchecked.
    - **Then:** The "Reservieren" button remains strictly disabled.
- **AC_09.2: Access to Information**
    - **Given:** The booking widget is active.
    - **When:** Clicking the "Datenschutzerklärung" link in the footer.
    - **Then:** Privacy policy opens in an isolated modal overlay without resetting the booking session.

US_10: Interface Accessibility (Accessibility - BITV 2.0)

- **AC_10.1: Keyboard Focus**
    - **Given:** Navigating through any step using the keyboard.
    - **When:** Pressing the Tab key.
    - **Then:** Interactive elements are focused sequentially with a high-contrast focus ring.
- **AC_10.2: Element Activation**
    - **Given:** An interactive element is focused.
    - **When:** Pressing Enter or Space.
    - **Then:** The target action executes correctly.
- **AC_10.3: Focus Trap in Modal Windows**
    - **Given:** A modal window (e.g., Datenschutz) is active.
    - **When:** Navigating via Tab key.
    - **Then:** Keyboard focus is trapped within the modal and closes on Escape key.
- **AC_10.4: Screen Reader Semantics**
    - **Given:** Dynamic components (accordions, counters) change state.
    - **When:** Component state changes.
    - **Then:** Corresponding ARIA attributes (aria-expanded, aria-disabled) update dynamically in the DOM.

System Resilience & Non-Functional Requirements (SYS_01 – SYS_05)

- **SYS_01 (AC_SYS_01): Session Timeout Warning (01:00)**
    - **Given**: The user is on Step 5 and session time elapses.
    - **When**: The countdown reaches 01:00 minute remaining.
    - **Then**: A warning modal appears allowing session extension.

- **SYS_02 (AC_SYS_02): Session Expiration & Hard Timeout (00:00)**
  - **Given**: The session warning modal expires without user interaction.
  - **When**: The countdown reaches 00:00.
  - **Then**: The session is annulled, form data is cleared, and the user is redirected to Step 1 (?rs).
- **SYS_03 (AC_SYS_03): Network Failure Handling (Offline Simulation)**
  - **Given**: Form entry is complete at Step 5.
  - **When**: Network connection drops during submission.
  - **Then**: DOM data is preserved and a network availability error is displayed.
- **SYS_04 (AC_SYS_04): Parallel Booking Conflict Handling**
  - **Given**: Concurrent requests are sent for a single appointment slot.
  - **When**: Two independent instances attempt booking simultaneously.
  - **Then**: The first request succeeds; the second thread intercepts Conflict 409 and displays a slot-occupied message.
- **SYS_05 (AC_SYS_05): Backend Failure / API 500 Error Handling**
  - **Given**: Transitioning from Step 3 to Step 4.
  - **When**: calendar.js returns HTTP 500 Internal Server Error.
  - **Then**: Navigation to Step 4 is blocked, displaying an explicit error message instead of an unhandled blank screen.

Infrastructure & Responsiveness Requirements (INF_01 – INF_03)

- **INF_01 (AC_INF_01): Mobile Layout Adaptability**
  - **Given**: Mobile context (iPhone / 430px viewport width).
  - **When**: Navigating through all booking steps.
  - **Then**: Zero horizontal scrolling occurs, and form inputs stack vertically into a single column.
- **INF_02 (AC_INF_02): Touch Target Sizing**
  - **Given**: Mobile interaction context.
  - **When**: Interacting with buttons, counters, and time tiles.
  - **Then**: Interactive elements meet the minimum target size of 44x44 pixels to prevent misclicks.
- **INF_03 (AC_INF_03): Keyboard Accessibility & Focus Control**
  - **Given**: Keyboard navigation mode across any step.
  - **When**: Pressing Tab, Enter, Space, or Escape keys.
  - **Then**: Focus order moves sequentially with a visible focus ring, and legal modals enforce focus traps.

1.6 Test Scenarios

1.6.1 Positive Scenarios

- **TS_01_A - E2E Flow: Bürgeramt**
  - **Description:** Execution of a positive end-to-end scenario from Step 1 through Step 5. Selecting the "Bürgeramt" department, exactly 1 service, a location in Homburg, and the first available time slot. Step 5 is completed with valid personal data.

- **Expected Result:** All steps up to the 5th are completed successfully; the "Reservieren" button transitions to the "enabled" state to proceed to Step 6.

- **TS_01_B - E2E Flow: Fahrerlaubnis**
  - **Description:** Execution of a positive end-to-end scenario from Step 1 through Step 5. Selecting the "Fahrerlaubnis" department, exactly 1 service, a location in Homburg, and the first available time slot. Step 5 is completed with valid personal data.
  - **Expected Result:** All steps up to the 5th are completed successfully; the "Reservieren" button transitions to the "enabled" state to proceed to Step 6.
- **TS_02 - UI: Service Counter Limits**
  - **Description:** Testing the business logic for service selection at Step 2. Verifying counter accuracy, "Weiter" button activation, and compliance with the system-wide limit (max 4 services). The test concludes at Step 2.
  - **Test Scenarios:**
    - **Initial State:** Verify that the "Weiter" button is in the "disabled" state when all counters are 0.
    - **Activation (0 → 1):** Increase the counter for any service to 1. Verification: "Weiter" button becomes "enabled", and the (-) button becomes active.
    - **Reset (1 → 0):** Decrease the service counter to 0. Verification: (-) button becomes "disabled", and the "Weiter" button reverts to "disabled".
    - **Reaching the Limit (3 → 4):** Increase the total count of services to 4. Verification: All (+) buttons on the page are automatically locked ("disabled"), while (-) buttons and "Weiter" remain active.
    - **Recovery from Limit (4 → 3):** Click the (-) button when 4 services are selected. Verification: All (+) buttons on the page are unlocked ("enabled"), and the "Weiter" button remains active.
  - **Expected Result:** All interface elements (counters, +/- buttons, "Weiter" button) synchronously toggle their states ("enabled"/"disabled") in strict accordance with the number of selected services, preventing exceeding limits or submitting an empty booking.
- **TS_03 - UI: Filter Interactivity**
  - **Description:** Proceeding to Step 4. Verifying accordion expansion, default date values, and the activation of time/day-of-the-week checkboxes. Testing filter application with dynamic availability processing. The test concludes at Step 4 or proceeds to Step 5 depending on appointment availability.
  - **Expected Result:**
    - The "Vorschläge filtern" accordion expands successfully; date range fields are pre-populated with the current date (From) and the date 21 days later (To), adjusted to the department's working days.
    - Time interval and day-of-the-week checkboxes are interactive and toggle to the "checked" state upon clicking.
    - **Outcome A (Slots Found):** The time grid is displayed; clicking the first available slot triggers the "Hinweis" modal window, and successful confirmation redirects the user to Step 5.

        - **Outcome B (No Slots Found):** A valid system message "Kein freier Termin verfügbar" (No free appointments available) is displayed; the test concludes at Step 4.
- **TS_04 - Navigation: Backward & Cache Stability**
    - **Description:** Verifying cache stability, preservation of the selected time slot, and dynamic data updates when navigating backward to Step 1.
    - **Expected Result:**
        - **Return to Step 4:** Navigating back from Step 5 to Step 4 retains: Funktionseinheit, Anliegen, Standort, and Termin. The Übersicht panel displays all four fields correctly.
        - **Return to Step 3:** Navigating back from Step 4 to Step 3 retains: Funktionseinheit, Standort, and Termin. The "Anliegen" data is cleared from the Übersicht panel.
        - **Return to Step 2:** Navigating back to Step 2 retains only: Funktionseinheit. All subsequent selections (Anliegen, Standort, Termin) are cleared from the Übersicht panel.
- **TS_14 - UI: Global Header & Overview Consistency**
    - **Description:** Verifying the stability and correct rendering of global interface elements (Header, Footer).
    - **Expected Result:**
        - **Header:** The logo and two icons are displayed on all steps (1–6); elements remain present and accessible throughout every step of the process.
        - **Footer:** All links (Hilfe, Impressum, Datenschutz, etc.) trigger modal windows upon clicking. The "Schließen" (Close) button within each window correctly closes the popup.
- **TS_15 - UI: Dynamic Übersicht & Step Indicator**
    - **Description:** Verifying the correctness of the dynamic "Übersicht" (Summary) block, data accuracy within it, and the synchronization of the Step Indicator with the current booking stage.
    - **Expected Result:**
        - **Step Indicator:** The progress indicator (1–5) visually highlights the currently active step corresponding to the page content.
        - **Übersicht (Dynamics):**
            - **Step 1:** The "Übersicht" block is hidden.
            - **Steps 2–5:** The block is present with the accordion expanded, displaying accurate information regarding selected services, location, and time.

1.6.2 Negative & Edge Case Scenarios

- **TS_06 - Filter: No Available Slots Handling**
    - **Status:** Blocked by Test Data. The test scenario is 80% automated (validation of the "Kein freier Termin" message).
    - **Reason:** Full implementation (filter reset + continuation) is blocked due to the inability to predict slot availability after filtering in the current test environment.

    - **Description:** A scenario where filter parameters at Step 4 return an empty result (no slots available). Verification of message display and filter reset via automated testing to continue the scenario.
    - **Expected Result:**
        - The calendar grid is hidden, and the message "Kein freier Termin verfügbar" (No free appointments available) is displayed. The filter is collapsed, and the "Zurücksetzen" (Reset) button is clickable.
        - The automated test clears the selected filter checkboxes, re-clicks "Filtern" (Filter), after which available slots appear.
        - The test clicks the first available slot, selects "Ja" (Yes) in the confirmation window, and proceeds to Step 5.
- **TS_07 - Security: Deep Linking Protection**
    - **Description:** Attempting a direct URL navigation to an internal booking step (Step 5) without an active session.
    - **Expected Result:** The system blocks access to the form and displays an error message.
- **TS_08 - Security: Form Validation**
    - **Description:** Verification of negative validation, input field security, and the state of the "Reservieren" (Reserve) button at Step 5 for both "Bürgeramt" and "Fahrerlaubnis" processes.
    - **Expected Result:**
        - Real-time Feedback: Upon entering invalid data (e.g., incorrect email format, invalid year, or mismatching emails), the system must immediately trigger a visual error response:
            - **Error Message:** Display of a specific red error text (e.g., "Die Mailadressen stimmen nicht überein").
            - **Visual Indicator:** The field/icon status changes from "valid" (green) to "invalid" (red).
        - **Button State Logic:** * The "Reservieren" button remains strictly disabled (`disabled` attribute / `aria-disabled="true"`) while mandatory fields are empty or contain invalid data.
            - The button becomes enabled **only after** all validation checks for mandatory fields are passed and the consent checkbox is checked.
- **TS_16 - System: Paste Restriction on Email Confirmation**
    - **Description:** Verifying that the email confirmation field ("Email wiederholen") strictly requires manual input by blocking the paste (Ctrl+V/Right-click) functionality.
    - **Expected Result:** The application must prevent any text from being pasted into the email confirmation field. The field must remain empty or unchanged upon a paste attempt, ensuring the user is forced to type the input manually. This restriction acts as a primary preventative measure before secondary validation (email mismatch) is triggered.
- **TS_09 - Session: Warning & Extension**
    - **Description:** Simulating browser clock progression via Playwright's `page.clock` to reach the 01:00 session threshold at Step 5, verifying client-side warning modal interaction and state persistence.

- **Expected Result:**
    - When the client-side session timer approaches the critical threshold (1 minute remaining), an "Ihre Sitzung läuft aus in 1 Minute" warning triggers on Step 5 and the warning modal opens.
    - Clicking the "Schliessen" button closes the modal window without resetting form inputs.
    - *Note:* This scenario validates client-side UI timer logic, warning overlay behavior, and form data retention under faked browser time. Server-side session renewal is outside the scope of virtual clock testing.
- **TS_10 - Session: Timeout Expiration**
  - **Description / Precondition:** The user is on Step 5 with a partially filled form. No interface interactions occur until the session timer (25 minutes) fully expires.
  - **Expected Result:** An automatic redirect to Step 1 occurs with a session reset parameter (URL: `.../homburg/?rs`). The Step 5 interface is completely locked, entered data is erased, and the footer timer restarts from the initial 25 minutes (displaying 24 on-screen).
- **TS_11 - System: Network Drop Simulation**
  - **Status:** Deferred.
  - **Reason:** Risk of data integrity violations in the live system. Forcing a connection drop during the booking submission process may result in "phantom" records or orphaned sessions in the database.
  - **Description:** The user is on Step 5 and has completed the form. At this moment, the browser context is switched to "Offline" mode. An attempt is made to submit the form or interact with elements requiring server communication.
  - **Expected Result:** The interface remains stable, and data in the DOM is preserved. Upon attempting to submit the form, the system displays a valid notification regarding the lost network connection (or the browser generates a standard network availability error, blocking the submission of "stale" data).
- **TS_12 - System: Parallel Booking Conflict**
  - **Status:** Deferred.
  - **Reason:** Test environment limitations (dependency on a shared backend). Executing parallel booking requests requires an isolated database with rollback capabilities after each run, which is not feasible without backend code intervention.
  - **Description:** Testing concurrent access to a single time slot by simultaneously sending requests from Step 5 to Step 6 using two independent Playwright instances.
  - **Expected Result:** The first request completes successfully (booking confirmed), while the second thread intercepts an API error (Conflict 409), and the interface displays a validation message to the user indicating that the selected slot is already occupied.
- **TS_13 - System: API 500 Error Handling**
  - **Description:** Verifying interface behavior when critical calendar data is unavailable (simulating an HTTP 500 error for `calendar.js`).

    - **Expected Result:** The system must handle the API 500 error gracefully. When calendar.js fails with HTTP 500 during transition from Step 3 to Step 4, navigation to Step 4 must be blocked, and the system should display a clear error notification ("Fehler bei der Terminauswahl") instead of rendering an unhandled blank screen. A blank screen or unhandled application freeze is considered an invalid UI state.
- **TS_17 - Error Message on Email Mismatch**
    - **Description:** Validation of the email confirmation logic by entering non-matching email addresses in the "E-Mail" and "E-Mail wiederholen" fields. The scenario proceeds through steps 1–4, then attempts to trigger validation by entering distinct email strings and shifting focus.
    - **Expected Result:** The system identifies the inconsistency and displays the specific error message: "Die Mailadressen stimmen nicht überein." in the designated error container.
- **TS_18 - System: Field Validation on Incomplete Submission**
    - **Description:** Verification of the form submission state when required fields in Step 5 are left incomplete. The scenario completes steps 1–4, navigates to step 5, and populates all mandatory personal data fields except for one (e.g., leaving the "Nachname" field blank).
    - **Expected Result:** The "Reservieren" button remains in a "disabled" state, preventing the user from proceeding to Step 6, as the validation requirements for mandatory fields are not met.

II: Test Execution & Strategy

- **2.1 UI/UX & Responsive Static Checklists**
- **2.2 Accessibility Auditing Checklist (BITV 2.0 / WCAG 2.1)**
- **2.3 Manual Test Cases**
- **2.4 Defect Reporting & Severity Criteria**
- **2.5 Definition of Done (DoD)**
- **2.6 Requirements Traceability Matrix (RTM)**
- **2.7 System Resilience & Interruption Scenarios**
- **2.8 Infrastructure & Accessibility Coverage**

2.1 UI/UX & Responsive Static Checklists

Block 1: Cross-Device Responsiveness (Mobile Context / Viewport 430px — iPhone 13/14)

Focus: Ensuring layout integrity, touch-target sizing, and interaction behavior on mobile screens (430px width).

[✓] **Vertical Input Layout (Step 5):** Verify that text fields (Vorname, Nachname) and the date-of-birth triad (Tag / Monat / Jahr) automatically stack into a single vertical column, preventing element overlapping.

[✓] **On-Screen Keyboard Interaction (Step 5):** Ensure that the activation of input fields (Vorname, Nachname, E-Mail) does not cause the layout to shift irreversibly, nor does the keyboard overlay the active field. The "Reservieren" button must remain accessible after the keyboard is dismissed.

[✓] **Touch Target Sizing:** Verify that navigation buttons (counters +/-, Step 4 time tiles) meet the minimum physical size of **44x44 pixels** to prevent accidental misclicks.

[✓] **Vertical Navigation Stack:** Ensure "Weiter" (Next) and "Zurück" (Back) buttons in the footer stack vertically on mobile, with the "Zurück" button positioned below the primary "Weiter" button.

[✓] **Horizontal Scroll Control:** Confirm the complete absence of horizontal scrolling across all 6 steps. Content must scale and adapt 100% to the 430px viewport width.

[✓] **Summary Panel (Übersicht) Behavior:** Verify that the "Übersicht" block remains consistently expanded and readable throughout all steps on mobile devices. Ensure that the block's content is clearly visible and does not overlap or obstruct any active input fields, particularly on Step 5.

Block 2: External Link Integrity & Modal Logic

Focus: Validation of footer links, modal window behavior, and session persistence.

[✓] **Modal Trigger:** Verify that clicking "Impressum" and "Datenschutz" in the footer triggers the respective modal overlay correctly (without reloading the page).

[✓] **Content & Closing:** Verify that the modal opens with the correct German text content and includes a functional "Close" button.

[✓] **External Links Logic:** Ensure that any links inside these modals (e.g., links to Homburg city pages) open in a new browser tab (`target="_blank"`).

[✓] **Session Integrity:** Verify that opening and closing these modals does not interrupt the booking process, reset the session timer, or clear selected services in the main tab.

Block 3: Brand Identity & Visual Hierarchy

[✓] **Action Buttons:** Verify that primary action buttons ("Weiter", "Reservieren") are displayed in dark-green, while secondary/grey buttons ("Abrechnen", "Zurück") are visually distinct.

[✓] **City Logo Stability:** Verify that the Homburg city logo is static. Clicking it should trigger no action, page reload, or session reset.

[✓] **Header Menus:** Verify that interactive header buttons (e.g., language selection) correctly open and close their dropdown menus without disrupting the booking session.

2.2 Accessibility Auditing Checklist (BITV 2.0 / WCAG 2.1)

Block 1: Keyboard Accessibility & Focus Handling

Focus: Verifying the capability to complete the entire booking process without the use of a mouse or trackpad.

[✓] **Tab Order Sequence:** Navigate through all 6 steps using the Tab key. Ensure focus movement is strictly sequential (top-to-bottom, left-to-right) across all interactive elements (buttons, accordions, inputs, time slots) without unexpected jumps or skipped elements.

[✓] **Focus Ring Visibility:** Confirm that every interactive element displays a clear, high-contrast focus ring when highlighted via keyboard navigation. The focus indicator must remain visible at all times across all steps.

[✓] **Element Activation (Enter / Space):** Verify that all primary actions—expanding accordions (Step 2), selecting a time slot (Step 4), and toggling the consent checkbox (Step 5)—execute correctly upon pressing Enter or Space.

[✓] **Modal Focus Trap:** When the "Datenschutzerklärung" (Privacy Policy) modal is triggered, ensure the keyboard focus is strictly contained within the popup. Tab navigation must cycle only through elements inside the modal (the "Hier" link, "Schliessen" button, and close icon) without leaking focus to the background UI.

[✓] **Escape Key Functionality:** Verify that the Privacy Policy modal closes instantly upon pressing the Escape key.

Block 2: Screen Reader Semantics & ARIA States

Focus: Ensuring interface accessibility for visually impaired users utilizing screen readers (e.g., NVDA, VoiceOver).

[✓] **ARIA Labels for Graphics:** Use the browser's developer tools (Inspector) to verify that all purely graphical elements—such as language switchers, close icons in modals, and field validation markers ([✓] or [!])—are equipped with descriptive `aria-label` attributes in German.

[✓] **Dynamic State Updates:** Verify that when an accordion is expanded, its state in the DOM tree updates to `aria-expanded="true"`, and when service limit buttons (+) are disabled, they update to `aria-disabled="true"`. This ensures screen reader software correctly announces these state changes to the user.

Block 3: Contrast & Interface Scaling

Focus: Verifying readability and usability for users with visual impairments.

[✓] **Text Contrast Ratio:** Use browser developer tools to verify that the contrast ratio between text (including fine print in the footer) and the background color is at least **4.5:1** (WCAG 2.1 Level AA standard).

[✓] **Page Zoom (200%):** Increase the browser zoom level to 200%. The interface must remain readable; text content must not overlap, and all interactive elements (including navigation buttons in the footer) must remain fully accessible and functional.

2.3 Manual Test Cases

- **TC_01: E2E Integration Test: Portal Entry Point**
    - **Acceptance Criteria (AC):** AC_01.1
    - **Priority:** High
    - **Preconditions:** Browser cache and cookies are cleared; no active booking session exists.
    - **Steps:**
        1. Navigate to the Homburg city portal: `[https://www.homburg.de/](https://www.homburg.de/)`.
        2. In the main navigation menu, click on "Rathaus".
        3. From the dropdown list, select "Stadtverwaltung", then navigate to "Online-Dienste".
        4. Locate the graphical tile labeled "Termin mit Bürgerbüro/Führerscheinstelle vereinbaren" and click it.
    - **Expected Result:** The widget's landing page (Step 1 of 6) loads successfully in a new browser tab. The address bar displays the target URL: `[https://termine-reservieren.de/termine/homburg/](https://termine-reservieren.de/termine/homburg/)`.
- **TC_02: Boundary Value Validation: Date of Birth Triad (Step 5)**
    - **Acceptance Criteria (AC):** AC_07.3, AC_07.4
    - **Priority:** High
    - **Preconditions:** Active session; user is on Step 5. Mandatory text fields (Vorname, Nachname, E-Mail, E-Mail-Wiederholung) are populated with valid test data. Privacy consent checkbox is checked. Date of birth fields (Tag, Monat, Jahr) are initially empty.
    - **Steps:**
        1. Clear the "Jahr" field, enter an unrealistic year (e.g., 1899), and unfocus the field.
        2. Verify the status marker next to the "Jahr" field and the state of the "Reservieren" button.
        3. Clear the "Jahr" field, enter a valid birth year (e.g., 1995), and unfocus.
        4. Verify the status marker next to the "Jahr" field and the state of the "Reservieren" button.
    - **Expected Result:**
        1. **Step 1–2 (Invalid Year):** The system identifies the year as being outside the valid range. A red error marker ([!]) appears next to the "Jahr" input, form validation fails, and the "Reservieren" button becomes disabled (cursor: not-allowed).
        2. **Step 3–4 (Valid Year):** A green success marker ([✓]) appears next to the "Jahr" field. The interface permits submission without Day/Month values; all mandatory validations are satisfied, and the "Reservieren" button becomes active.

- **TC_03: UX Isolation & Legal Modal Logic**
    - **Acceptance Criteria (AC):** INF_02
    - **Priority:** Medium
    - **Preconditions:** User is on Step 1 of the booking widget. The global session timer in the footer is active and counting down.
    - **Steps:**
        1. Scroll to the bottom of the page and click the "Datenschutzerklärung" link in the footer.
        2. Visually confirm the appearance of the modal window, the correctness of the German text, and the background overlay (blocking background interaction).
        3. Locate the "Hier" hyperlink within the modal text (linking to the full DSGVO document) and click it.
        4. Return to the initial widget tab and click the "Schliessen" button (or the close [X] icon in the top-right corner).
    - **Expected Result:**
        1. **Step 2:** The legal information modal opens as an overlay. The background is dimmed, and interactions with the Step 1 department selection buttons are completely disabled.
        2. **Step 3:** The full privacy policy document opens in a new browser tab (`target="_blank"`). The booking session and the modal window on the original tab remain active and state-preserved; the footer timer continues to count down without interruption.
        3. **Step 4:** The modal window closes smoothly. The user is returned to Step 1 in its initial state, and no previously selected parameters are reset.
- **TC_04: Input Validation for 'Jahr' with Special Characters Acceptance**
    - **Criteria (AC):** AC_07.3 (Date of Birth Validation Logic (Geburtsdatum))
    - **Priority:** High
    - **Preconditions:** Active session; user is on Step 5. Mandatory personal data fields are filled with valid test data. Privacy consent checkbox is checked.
    - **Steps:**
        - Navigate to Step 5 (Personal Data).
        - Locate the "Jahr" (Year) input field.
        - Enter a valid year followed by a special character (e.g., "1995,").
        - Click outside the input field to trigger the blur/validation event.
        - Observe the validation status marker next to the "Jahr" field and the state of the "Reservieren" button.
    - **Expected Result:**
        - The system must sanitize the input and reject any non-numeric characters.
        - A red error marker ([!]) must appear next to the "Jahr" field.
        - The specific error message "Bitte geben Sie ein zulässiges Geburtsdatum ein" must be displayed.
        - The "Reservieren" button must remain disabled.

    - **Actual Result (Bug BR_01):** The system erroneously accepts the input as valid, displays a green success marker ([✓]), and enables the "Reservieren" button, potentially allowing data corruption in the database.

2.4 Defect Reporting & Severity Criteria

Defects discovered during manual or automated testing are classified by their impact on business logic (Severity) according to the following criteria:

Severity	**Technical Criteria**	**Real-world Example (Homburg.de)**
Blocker	Total failure of the critical path (Happy Path). No workaround available. UI results in "White Screen of Death" or infinite loading.	HTTP 500 error on Step 1. The "Weiter" button on Step 2 is unresponsive, making booking impossible.
Critical	Violation of core business logic or security. Essential features are blocked, despite potential complex workarounds.	The system allows form submission on Step 5 without checking the GDPR consent box. The session timer terminates prematurely.
Major	Functionality is operational but with significant deviations from UI standards or logic. The core business path remains viable.	Step 4 calendar displays/allows selection of dates from the previous month due to layout bugs. Filter reset function is fully broken.
Minor	Superficial UI defects, typos, or slight deviations from design mockups that do not impact the booking flow.	Grammatical error in a German label (e.g., lowercase "vorname"). The success icon [✓] is misaligned by 2 pixels.

Defect Reporting Workflow

When a defect is discovered that is not covered by automated tests, a formal Bug Report must be created. To ensure efficiency in the resolution process, every report must include the following elements:

- **Unique ID:** A reference identifier for tracking.
- **Summary:** A concise description of the issue.
- **Steps to Reproduce:** A step-by-step walkthrough to trigger the defect.
- **Actual Result:** A description of what is currently occurring.
- **Expected Result:** A description of the desired system behavior.
- **Mandatory Artifacts:** Evidence such as UI screenshots, screen recordings, or browser console logs (Trace Viewer logs for automated failures).

2.5 Requirements Traceability Matrix (RTM)

Requirement ID	Acceptance Criterion	Description	Test Case / Method	Status / Coverage
US_01	AC_01.1	Successful page load of the landing page	TC_01	Covered (Manual)
US_01	AC_01.2	Deep linking protection	TS_07	Covered (Auto)
US_02	AC_02.1	Automatic navigation to Step 2 upon department selection	TS_01_A/B	Covered (Auto)
US_03	AC_03.1	Accordion UI behavior (expand/collapse)	TS_02	Covered (Auto)

US_03	AC_03.2	Counter Plus & Navigation Activation (0 $\to$ 1)	TS_02	Covered (Auto)
US_03	AC_03.3	Counter Minus & Navigation State (1 $\to$ 0)	TS_02	Covered (Auto)
US_03	AC_03.4	Service Limit Enforcement (Locking (+) buttons)	TS_02	Covered (Auto)
US_03	AC_03.5	Service Limit Enforcement (Unlocking (+) buttons)	TS_02	Covered (Auto)
US_04	AC_04.1	Required documents modal overlay rendering	TS_14	Covered (Auto)
US_05	AC_05.1	Branch display & interactive map rendering	Exploratory / Manual	Exploratory / Manual
US_05	AC_05.2	Popup activation on branch marker-pin click	Exploratory / Manual	Exploratory / Manual
US_05	AC_05.3	Location selection and external	Exploratory / Manual	Exploratory / Manual

		Google Maps navigation		
US_06	AC_06.1	3-Week horizon availability calendar display	TS_03	Covered (Auto)
US_06	AC_06.2	Filter interaction (Accordion filter apply & reset)	TS_03	Covered (Auto)
US_06	AC_06.3	No available slots error scenario handling	TS_06	Covered (Auto)
US_06	AC_06.4	Time selection "Hinweis" modal window trigger	TS_03	Covered (Auto)
US_07	AC_07.1	Global validation UI mechanics (indicators [!]/[✓])	TS_08	Covered (Auto)
US_07	AC_07.2	Cross-field email verification & anti-paste restriction	TS_16	Covered (Auto)
US_07	AC_07.3	Date of Birth validation logic (Valid year input)	TC_02, TC_04	Covered (Manual)

US_07	AC_07.4	Submit button lock state when mandatory fields are empty	TS_08	Covered (Auto)
US_07	AC_07.5	Submit button activation when all fields are valid	TS_08	Covered (Auto)
US_07	AC_07.6	Backward Navigation & cache stability across steps	TS_04	Covered (Auto)
US_07	AC_07.7	Dynamic Übersicht block & Step Indicator sync	TS_15	Covered (Auto)
US_08	*N/A*	Booking Confirmation (Step 6)	Out of Scope	Out of Scope
US_09	AC_09.1	GDPR compliance mandatory checkbox interface blocking	TS_08	Covered (Auto)
US_09	AC_09.2	Isolated privacy policy modal window access	TC_03	Covered (Manual)
US_10	AC_10.1–10.4	Keyboard Navigation & ARIA States Audit	Checklist 2.2	Covered (Manual)

2.6 System Resilience & Interruption Scenarios

Requirement ID	Acceptance Criterion (AC)	Description	Test Case / Method	Status
SYS_01	AC_SYS_01	Session timeout warning (1:00 min)	TS_09	Covered (Auto)
SYS_02	AC_SYS_02	Session expiry & redirect (0:00)	TS_10	Covered (Auto)
SYS_03	AC_SYS_03	Network failure (Offline mode)	TS_11	Deferred
SYS_04	AC_SYS_04	Parallel booking (Race condition)	TS_12	Deferred
SYS_05	AC_SYS_05	Backend failure / API 500 error	TS_13	Covered (Auto)

2.7 Infrastructure & Accessibility Coverage

Requirement ID	Acceptance Criterion (AC)	Description	Test Case / Method	Status
INF_01	AC_INF_01	Mobile layout & touch targets (iPhone)	Checklist 2.1 (Block 1)	Covered (Manual)

INF_02	AC_INF_02	Touch target sizing (44x44px minimum)	Checklist 2.1 (Block 1)	Covered (Manual)
INF_03	AC_INF_03	Keyboard navigation (Tab order)	Checklist 2.2 (Block 1)	Covered (Manual)

2.8 Definition of Done (DoD)

2.8.1 Manual Testing Done

[✓] **Requirement Coverage:** All User Stories (US_01–US_10) have been broken down according to business requirements; logical nuances (e.g., the date of birth triad) are documented; and each User Story is equipped with BDD-formatted (Given-When-Then) acceptance criteria that eliminate ambiguity and describe verifiable system behavior.

[✓] **Test Design:** Manual checklists have been designed for visual UI/UX inspections and responsiveness verification across mobile devices.

[✓] **Accessibility:** A manual interface audit has been conducted based on the BITV 2.0 standard (specifically verifying keyboard navigation using Tab, Enter, and Space).

[✓] **Traceability:** The Requirements Traceability Matrix (RTM) is complete, confirming that no business requirement remains untested.

[✓] **Risk & Resilience Analysis:** Negative scenarios and critical business risks (session timeouts, race conditions, network drops, and deep-linking protection) have been integrated into the testing architecture.

2.8.2 Automation Testing Done

[✓] **Code Coverage:** Automated functional and system scenarios have been fully implemented using Playwright and **JavaScript (Node.js)**.

[✓] **Calendar Stability:** Flexible logic for identifying available dates has been implemented, ensuring tests remain stable when the calendar month rolls over (Anti-Flaky Logic).

[✓] **Production Isolation:** E2E automation validates the user flow from Step 1 to Step 5. Step 6 is explicitly excluded from execution to prevent submitting dummy data into the live Homburg production database.

[✓]**Cross-Browser Compatibility:** Tests pass consistently (Green Build) in parallel mode across three browsers: Chromium, Firefox, and WebKit.

[✓] **Execution Stability:** In the event of a test failure, the framework automatically generates artifacts (screenshots and Trace Viewer logs) to facilitate rapid debugging.

[✓] **Sync & Review:** The requirements document has been cleared of implementation code, reviewed by the QA Lead, and approved as the "Single Source of Truth" for both manual and automated testing.

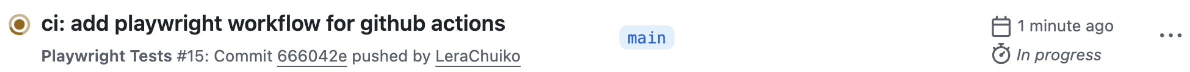
III. Test Automation Framework Engineering & Scripting

Focus: Designing a scalable framework based on Playwright + JavaScript. Development of robust E2E tests for positive, negative, and system scenarios utilizing API mocking.

3.1 Technology Stack Rationale

The booking widget (SUT) requires high-speed execution and deep network control. The Playwright + JavaScript stack was selected for the following reasons:

- **Architecture & Speed:** Unlike Selenium, Playwright interacts with browsers via internal debugging protocols (CDP), enabling near-instant clicks and DOM interactions without the overhead of a WebDriver proxy.
- **Built-in Auto-waiting:** Playwright inherently validates element visibility and clickability, eliminating the need for flaky hardcoded pauses (e.g., `setTimeout`).
- **Browser Context Isolation:** The ability to generate isolated `BrowserContexts` per test ensures zero cross-test interference, crucial for session-dependent booking flows.
- **Native Network Interception:** With built-in `page.route()`, the framework mocks API responses (booking success, scheduling slots, backend failures) directly without third-party tools.
- **JavaScript Efficiency:** Using a native Node.js/JavaScript environment ensures seamless integration with modern CI/CD pipelines and low barrier for maintenance.



3.2 Directory & Project Structure

The project follows a streamlined structure designed for clarity and rapid maintenance. All functional test scripts, data definitions, and configuration files are consolidated within the `tests/` directory to simplify navigation and project growth.

```
.
├── .github/
│   └── workflows/
│       └── playwright.yml          # CI/CD configuration manifest for GitHub Actions
├── assets/
│   ├── E2E_playwright.mp4          # Video recording of the automated end-to-end flow
│   ├── Test_execution_report_pw.png  # Graphic screenshot of local Playwright test run
│   └── UI_screenshots.pdf          # Compiled documentation of interface elements layout
├── docs/
│   ├── Bug_report.pdf              # Formal documentation for discovered application defects
│   ├── Checklist.pdf               # Comprehensive manual UI/UX and responsiveness log
│   ├── Manual_test_cases.pdf       # Step-by-step conditions and expected results ledger
│   ├── Project_overview_test_plan.pdf# Master project strategy blueprint Document
│   ├── Test_execution_report.pdf   # Log of processed scenario verdicts and behaviors
│   └── Test_scenarios.pdf          # High-level architecture map of mapped functionalities
├── tests/
│   ├── e2e_booking.spec.js         # Full end-to-end reservation workflow validation scripts
│   ├── helpers.js                  # Shared UI utilities and interactive execution shortcut tools
│   ├── session_timeout.spec.js     # Scripts validating session hold rules and warning prompts
│   ├── system_errors.spec.js       # Assertions tracking application stability on API failures
│   ├── system_resilience.spec.js   # Validations ensuring smooth performance over disruptions
│   ├── testData.json               # Static dictionary of clean input values and field parameters
│   ├── ui_components.spec.js       # Component tests checking counters, maps and accordions
│   └── validation_security.spec.js # Automated scripts blocking text pasting and form tampering
├── .gitignore                      # Standard configuration defining untracked repository patterns
├── LICENSE                         # Project distribution legal licensing file framework
├── README.md                       # Main repository entry guide and configuration manuals
├── package-lock.json               # Auto-generated file locking exact node dependency states
├── package.json                    # Master script automation definitions and package metadata
└── playwright.config.js            # Central engine runtime configurations, contexts and timeouts
```

3.3 Architecture Note: Helper-Based Design

The framework utilizes a **Helper-Based Modular Architecture** rather than a standard Page Object Model.

- **Rationale:** By modularizing common UI actions into discrete, reusable helper functions (e.g., `step_1_SelectDepartment`, `step_5_FillForm`), the framework achieves high agility.
- **Logic Isolation:** Locators are centralized in the top-level helper file, while complex business logic (like date calculations in `getFormattedFutureDate` or state verification in `verifyStepIndicator`) is encapsulated in functional helpers.
- **Maintenance:** This design allows for rapid modification of booking steps. Since the helpers are decoupled from specific test files, a single update in the helper library propagates across all test suites, ensuring high maintainability without the overhead of class-based POM structures.

3.4 Data Injection & Dynamic State Control

Instead of relying on real-time server data, which is often unpredictable, the framework utilizes **DOM-level data injection** to verify UI components and system resilience.

- **Deterministic UI Verification:** In scenarios like TS_05 (Predictive Slot Display), the system renders dynamic information (e.g., "Nächster Termin"). To ensure the test is deterministic, we use **page.evaluate()** to inject test-specific data (e.g., a calculated future date) directly into the UI node, allowing us to test visual states even when live slots are unavailable. In parallel, for environment simulation like TS_13 (API 500

Error), the framework uses **page.route()** to mock network-level responses, combined ensuring comprehensive control over both data injection and API behavior.
- **Security & Sanitization Validation:** We utilize data injection to perform negative testing, such as inserting XSS payloads into input fields. This allows us to confirm that the application's sanitization logic correctly identifies and blocks unauthorized input, maintaining system integrity.
- **Benefits of DOM Injection:**
    - **Independence:** Tests do not fail due to external backend availability or "no slots" statuses at the city office.
    - **Precision:** We can force specific, edge-case date formats or long strings into the UI to verify that the layout remains responsive and does not break.
    - **Verification:** This technique allows us to test the rendering logic of the application, ensuring that the UI correctly handles and displays whatever data the backend provides.

IV. CI/CD: Implementation Status and Constraints

- **Objective:** I configured a CI/CD pipeline using GitHub Actions to automate test execution on every commit.
- **Challenge:** During implementation, it became clear that the target website (`termine-reservieren.de`) actively blocks requests from cloud environments (like GitHub Actions), identifying them as suspicious bot activity.
- **Outcome:** Automated execution in the CI/CD environment is currently unstable due to the target site's security restrictions.
- **Resolution:** To ensure reliable results, the primary testing process is currently maintained locally, where network conditions allow for the full execution of the test suite. This approach avoids "flaky" or false-negative results and allows me to focus on maintaining high code quality rather than bypassing external security layers.

V. Maintenance & Test Stability

- **Easy Maintenance:** I keep the code clean by using helper functions to separate test logic from UI selectors. If the website design changes, I only update one file instead of fixing dozens of tests.
- **Built-in Stability:** I rely on Playwright's auto-waiting features instead of manual pauses (`sleep`), which makes the tests faster and more reliable.
- **Smart Locators:** I prioritize robust locator strategies, utilizing semantic locators (getByRole, getByLabel) where available, alongside stable IDs and resilient CSS selectors (#WeiterButton, .btn-number) to ensure long-term test stability.
- **Root Cause Analysis:** When a test fails, I use Playwright Trace Viewer to see exactly what happened, allowing me to fix the underlying issue quickly rather than just guessing.